CIS 2450 — BIG DATA ANALYTICS — SPRING 2026

Predicting Song Genre, Danceability, and Energy

From lyrics, Spotify audio features, and Deezer metadata

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FINAL PROJECT

OBJECTIVE

Making three predictions

Streaming services like Spotify, Apple Music, etc. need to auto-tag millions of uploaded tracks.

Learned classification scales far more efficiently than manual review.

Our project predicts a song's broad genre, danceability, and energy from three sources: lyrics, Spotify-derived audio features, and Deezer metadata.

GENRE

10-way classification
(Rock → Classical)

DANCEABILITY

Regression
(0 — 1 scale)

ENERGY

Regression
(0 — 1 scale)

DATASET

433,618 songs from a Kaggle x Deezer inner join

498,978

Spotify source rows

After deduplication of Kaggle 550K+

86.9%

match rate

After Deezer API enrichment

433,618

final corpus

Cleaned, joined, in DuckDB

EXTRACT & DEDUPLICATE

Kaggle CSV -> *songs_raw* -> deduplication -> songs_no_duplicates

›

REQUEST

Deezer API -> *deezer_data*

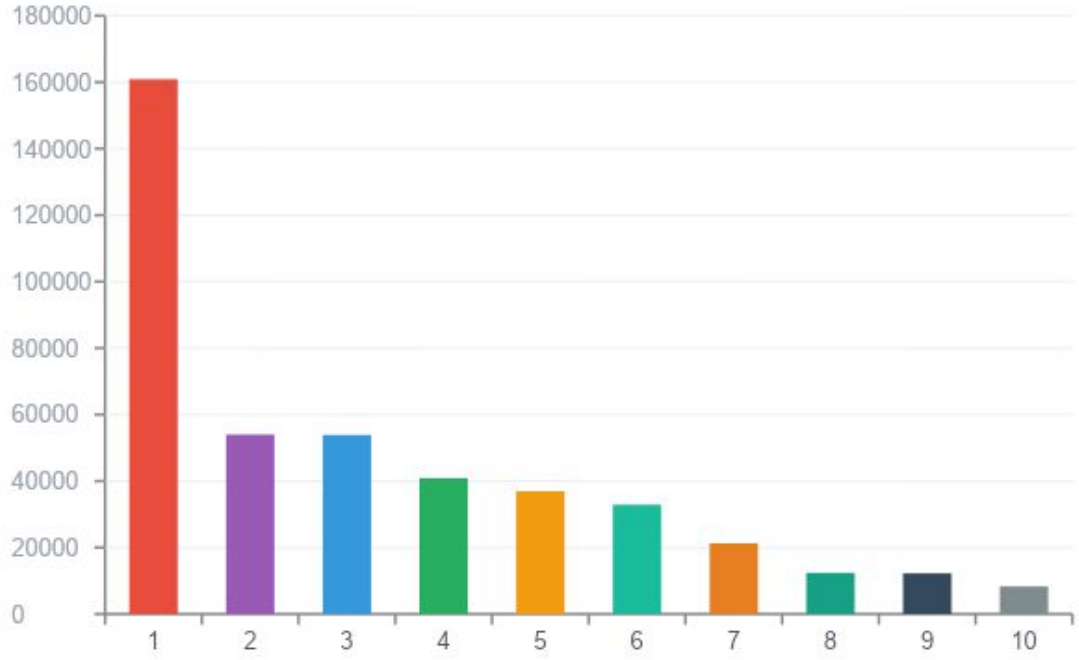
›

INNER JOIN

songs_no_duplicates + deezer_data -> inner join on *id* -> *songs* -> *songs.db*

Three SQL stages: deduplicate songs_raw -> fetch deezer_data -> INNER JOIN on _id

19× class imbalance shapes the metric choice



37.1%

always-Rock baseline
any model must clear this

10%

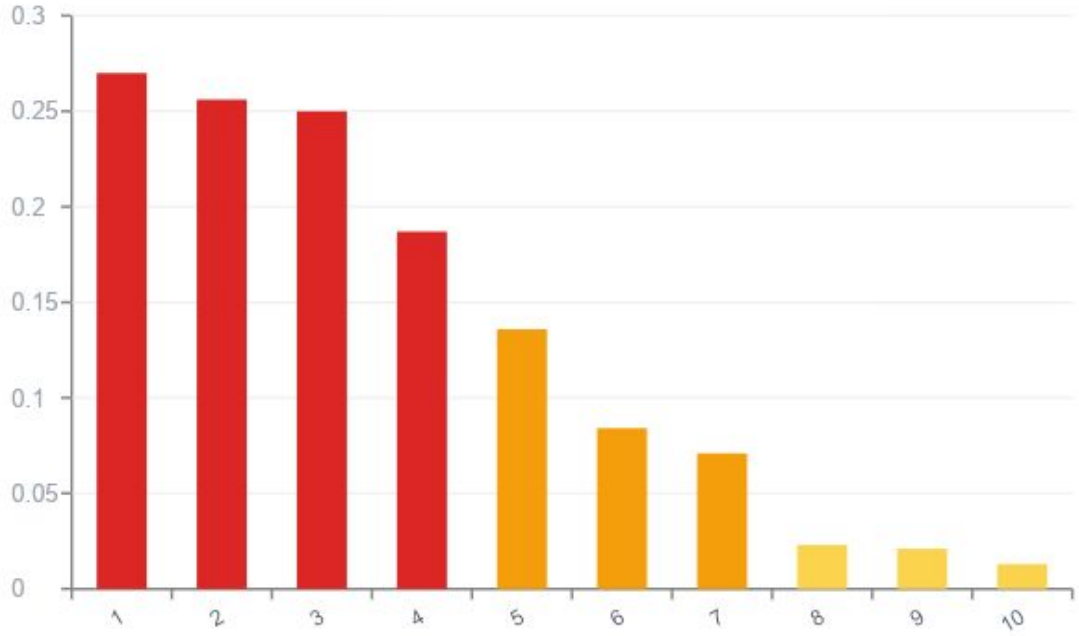
random baseline
(uniform over 10 classes)

MACRO F1

our primary metric
weighs every genre equally

EDA — FINDING 2

Four audio features carry the signal (ANOVA η^2)



EFFECT SIZE (η^2)

- Large (≥ 0.14)
- Medium (0.06–0.14)
- Small (0.01–0.06)
- Negligible (< 0.01)

KEY TAKEAWAY

Danceability, energy, acousticness, and speechiness all show large effect sizes.

Among the audio features, these same four rank as the model's top four most-important — EDA called the shots within this block.

MODELING PIPELINE

From raw song to a 521-dimensional vector

DISCIPLINE: stratified 80/20 train-test split is the FIRST transformation. Every preprocessor below fits on training data only.

521-dim feature vector ⟶ fed to all classifiers and regressors

Numeric audio

RobustScaler
10 Spotify features + lyric word count

Lyrics (LSA)

TF-IDF (10K features, sublinear)
⟶ Truncated SVD

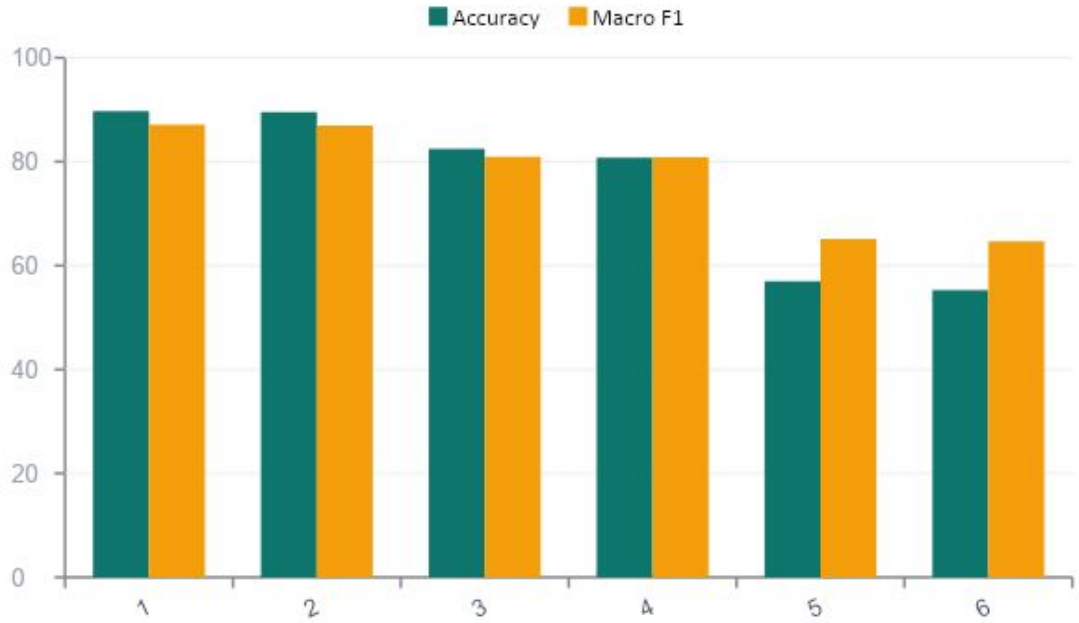
K-Means cluster

k=10 fit on scaled audio
ID one-hot encoded

Niche tag one-hot

Top-200 sub-genre tags
(88.7% song coverage)

LR won. Linear models led the field.



WINNER

87.1%

macro F1 — LR baseline on held-out test set

RF > ADABOOST

+16 pts

macro F1 gap.
Bagging beat boosting.

TUNING DIDN'T HELP LR

−0.21 pts

RandomizedSearchCV found a flat plateau.

MODELING — REGRESSION

Energy is highly predictable. Danceability is not.

ENERGY

$R^2 = 0.823$

RMSE 0.103

Strong correlations with loudness and acousticness drive a high R^2 — the model recovers energy almost from physics alone.

DANCEABILITY

$R^2 = 0.571$

RMSE 0.113

No single dominant linear correlate. Lyrics provide partial signal but the ceiling is much lower.

Both winners: Ridge regression. *Target column dropped from input to prevent trivial leakage — model recovers the target from lyrics + the OTHER audio features.*

INSIGHTS

What we learned from the comparison

01 EDA Dominated

ANOVA's top-four audio features ended up as the Random Forest's top-four most-important audio features. Investing in EDA up front made modeling confirmatory rather than exploratory.

02 Simple won

Logistic Regression beat both Random Forest and AdaBoost. Once you've engineered 521 informative dimensions, a linear boundary is enough — compute is no substitute for representation.

03 Bagging vs boosting revealed a problem structure

Random Forest beat AdaBoost by 16 macro-F1 points. That gap told us the error is variance-dominated, not bias-dominated. A useful diagnostic, not just a horse race.

Where the project ends and what comes next

LIMITATIONS

Coarse genre labels.
Pop ↔ Electronic confusion is partly genuine label ambiguity.

Predicting model outputs.
Spotify's audio features are themselves model outputs. Raw audio embeddings would be stronger.

SMOTE on binary features.
Linear interpolation across 200 binary tag columns produces semantically odd synthetic songs. SMOTENC would be more principled.

FUTURE WORK

Raw audio embeddings.
Replace Spotify's hand-engineered features with CLAP or MERT to push regression R^2 higher.

Multi-label genres.
Treat niche tags as multi-label targets — closer to how playlists actually generate.

Per-decade models.
1980s Rock and 2010s Rock sound different. Per-era models would let us study genre drift.